INCH-POUND
MIL-DTL-81940/4B
7 October 2011
SUPERSEDING
MIL-V-0081940/4A(SH)
25 February 1992
MIL-V-81940/4
14 December 1990

DETAIL SPECIFICATION SHEET

VALVE, SAMPLING AND BLEED, HYDRAULIC, 3000 PSI NON-AIRCRAFT APPLICATIONS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-V-81940.

REQUIREMENTS:

1. The valve (see figure 1) shall be actuated by a rotary type knob or pushbutton. Release of the knob or button shall automatically return the valve to the closed position. The rotary type knob shall be fully opened with a maximum rotation of 180 degrees. The valve knob or button return mechanism shall be independent of the valve poppet, allowing the valve poppet to close automatically by fluid pressure upon release or failure of the knob or button return mechanism.

2. Torque required to actuate the rotary knob type valve to a fully open position shall not exceed 10 inch-pounds with an operating pressure of 3,000 pounds per square inch (lb/in^2) applied to the valve inlet.

3. Force required to actuate the pushbutton type valve to a fully open position shall not exceed 25 pounds with an operating pressure of 3,000 lb/in^2 applied to the valve inlet.

4. Valve shall have a protective cap with secondary sealing provisions. Protective cap shall include a security chain, or equivalent, to prevent loss of cap.

5. The part number shall consist of the basic number of this specification sheet plus a dash number indicating end fitting and spout as specified in table I.

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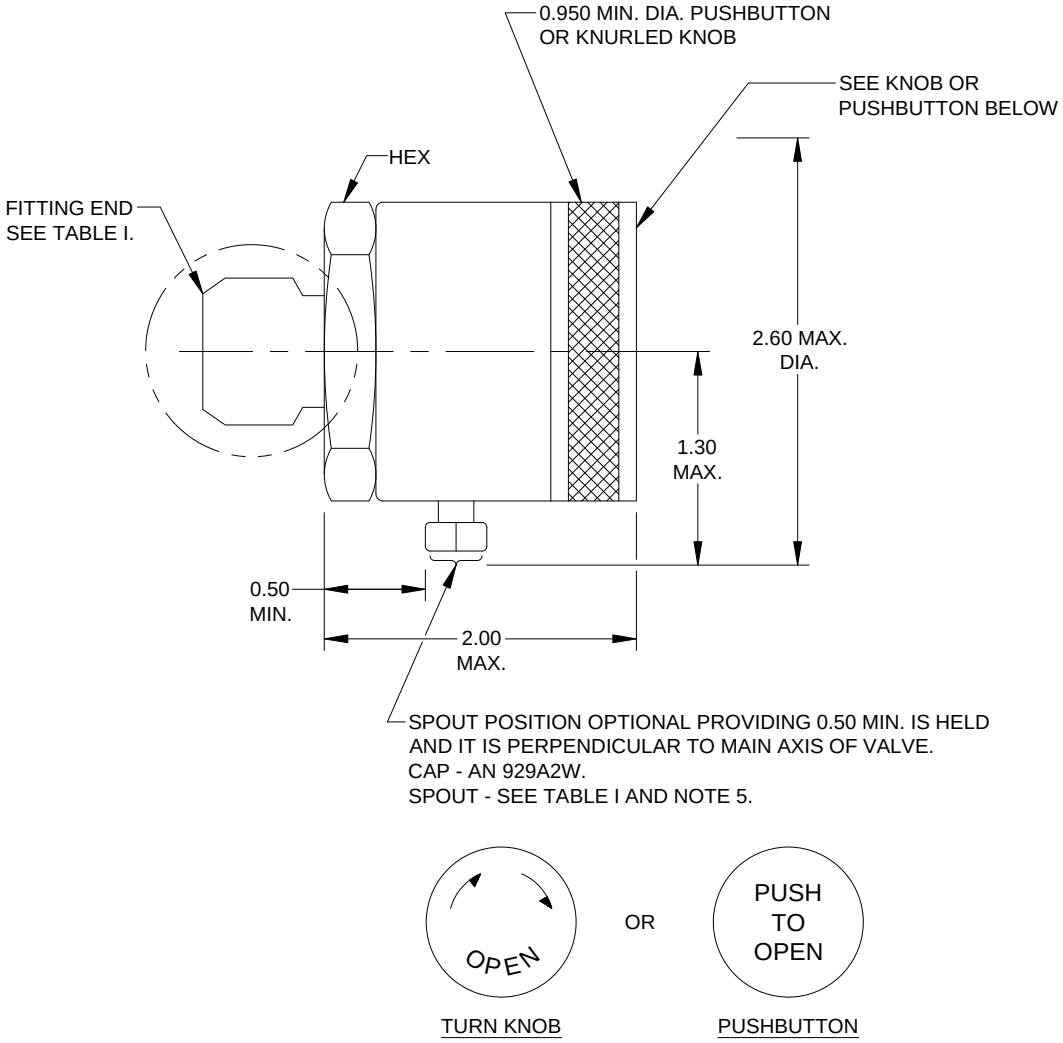

NOTES:

1. Dimensions are in inches.
2. Tolerances: Decimals ±0.010, angles ±1 degree.
3. Surface roughness shall not exceed 125 microinches in accordance with ASME B46.1. All burrs shall be removed and sharp edges broken 0.005 inch minimum.
4. Material shall be in accordance with MIL-V-81940.
5. The spout shall be in accordance with table 1, except the bore may be up to 0.103 inch in diameter.

FIGURE 1. Valve.

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TABLE I. Valve construction.

Part no.	Fitting End	Spout End
M81940/4-1	SAE-AS4395 E 04 or 37-degree flare tube fitting end for ¼ inch nominal tube outer diameter (OD) fitting shown by figure 3B of SAE-J514	SAE-AS4395 G 02
M81940/4-2	Tapered pipe thread end for ¼ inch nominal tube OD fitting shown by figure 2A of SAE-J514	SAE-AS4395 G 02
M81940/4-3	Tapered pipe thread end for ¼ inch nominal tube OD fitting shown by figure 2A of SAE-J514, except that thread is specified as ¼-18 NPT in accordance with FED-STD-H28 or ANSI/ASME B1.20.1	SAE-AS4395 G 02

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army – AV
Navy – SH
Air Force – 11

Review Activity:
Air Force – 71
Navy – AS
Navy – MC
Navy – SA

Preparing Activity:
Navy – SH
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NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil.

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